Teaching Notes 2B
The Capital Asset Pricing Model (CAPM)

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Overview

- The Capital Asset Pricing Model (CAPM)
- Firm-Specific Risk vs. Market Risk
- Estimating Betas using Regression Analysis

Objectives

- Calculate risk-adjusted returns in order to value both financial securities and investment projects.
- Understand the Capital Asset Pricing Model (CAPM).
- Learn how to estimate betas and compute the cost of equity capital of a project.

The Capital Asset Pricing Model (CAPM)

Introduction

- CAPM represents an idealized view of how the market prices securities and determines **expected returns**.
  - Expected returns are determined in equilibrium by risk.
  - Risk is a function of variances and covariances.

- Using historical average returns for individual assets is a very noisy guide to expected returns.
  - Instead, we need to estimate **betas**, which are a measure of market risk, and plug them into the CAPM equation to get expected returns.

The CAPM Formula

- This CAPM states that:
  - If all investors prefer high return and low risk (= variance)
  - If all investors have same information and no market frictions
- Then in equilibrium the expected return on risky asset i is:

$$\boldsymbol{E(r_i) = r_f + \beta_i[E(r_m) - r_f]}$$

where:

- $E(r_i)$ = Expected return of asset i; r_f = risk-free rate
- $\beta_i = \frac{Cov(r_i, r_m)}{Var(r_m)} = \frac{Cov(r_i - r_f, r_m - r_f)}{Var(r_m - r_f)}$ = measure of risk of asset i;
- $E(r_m)$ = Expected return of market portfolio
- $[E(r_m) - r_f]$ = Market Risk Premium ($\approx$ 7%)

The Capital Asset Pricing Model: Intuition

- Investors hold a security in their portfolio if it provides **extra reward** (*i.e.* expected return) **for the risk** it adds to the portfolio.
  - It adds expected excess return ($E(r_i) - r_f$).
  - It adds risk **proportional** to its **beta** with the portfolio (*i.e.* covariance).

- CAPM tells you that what a security adds to the risk of the portfolio will be just offset by what it adds in terms of expected returns.
  - Securities with more risk must also promise a higher expected excess return.
  - In equilibrium, investors should be indifferent between investing across assets => ratio of **marginal return** to **marginal risk** must be the same for all assets.

CAPM Intuition

- **The risk of each stock is measured by its "beta":**
  - Stocks with beta > 1 increase the volatility of a diversified portfolio, and stocks with beta < 1 decrease the volatility of a diversified portfolio

- This implies that in equilibrium all stocks must earn the <u>same</u> extra compensation of risk per unit of risk taken. For all stocks:

$$\boldsymbol{\frac{E(r_A) - r_f}{\beta_A} = \frac{E(r_B) - r_f}{\beta_B}}$$

  - If this wasn't true, say A earns higher extra return than B, then investors would all buy A and drop B, and the price of A would increase and price of B would drop. This would increase the expected return of A and decrease the one of B, until the equilibrium is re-established.

- The equality must hold for the market too. That is, for all stocks:

$$\frac{E(r_A) - r_f}{\beta_A} = \frac{E(r_m) - r_f}{\beta_m}$$

- The beta of the market is 1: $\beta_m = 1$, yielding the CAPM.

Using the CAPM

- What inputs do we need?
  - We need three “things”: β, r_f, and $(r_m - r_f)$!

- Let’s see where we can get these:
  - r_f is chosen based on the available yield curve and its maturity matches the horizon of the investment.
    - Example, the spot interest rate on T-Bills
  - Market Risk Premium: $(r_m - r_f) = 7\%$.
    - Based on historical estimates
  - β is computed based on a “regression” of excess firm stock returns on excess market returns

Using the CAPM

- The market portfolio is the portfolio of all risky securities held in proportion to their market value.
- This means that the return on the market portfolio is given by:

$$r_m = \sum_{i=1}^{N} x_i^v \times r_i$$

where: $x_i^v = \dfrac{\text{Total Dollar Value (Market Capitalization) of Security i}}{\text{Total Dollar Value (Market Capitalization) of all Risky Securities}}$

- **Note:** The market portfolio should include **all** securities that investors hold, not just stocks (i.e. bonds, real estate, human capital).
  - However, in practice, practitioners use the market portfolio of US stocks as a market proxy: **The S&P 500 Index.**

Using the CAPM

Beta and CAPM returns for different stocks

Stock	Average Return (2019 - 2024)	Standard Deviation (2019-2024)	Beta (2019 - 2024)	CAPM Expected Return Er = rf + beta * MRP (rf = 5.47% and MRP = 7%)
Amazon	17%	33%	1.15	14%
Apple	31%	30%	1.24	14%
Microsoft	25%	22%	0.89	12%
Nvidia	75%	49%	1.61	17%
Ford	16%	45%	1.65	17%
GM	11%	40%	1.46	16%
Tesla	75%	77%	2.33	22%
Pfizer	-2%	28%	0.69	10%
Merck	13%	21%	0.41	8%
Exxon Mobile	18%	34%	0.89	12%
Coca Cola	7%	19%	0.61	10%
Walmart	13%	18%	0.49	9%

Using the CAPM

- **Bottom line 1:** You can use the CAPM to find the discount rate for a new investment such as a capital investment project.

- **Example:**
  - Suppose we are analyzing a proposal by Pfizer to expand its capacity.
  - According to the previous table, investors are looking for a return of 10% from businesses with the risk of Pfizer.
  - So the cost of **equity** capital for a further investment in the same business is 10%.

Using the CAPM

- **Bottom line 2:** You can use the CAPM as benchmark to find investment opportunities that yield higher “risk-adjusted” returns.

- **Example:**
  - From 1990 to 2023, value stocks, i.e. stocks with low Market-to-Book ratio, had an average excess return of 10.5%/year.
  - Their beta was 1.09 and the MRP was 8.4%.
  - According to the CAPM, the average return should have been

$$E[r_{value}] - r_f = \beta_{value} \times MRP = 1.09 \times 8.4\% = 9.16\%$$

  - Value stocks yielded a higher excess return compared to the risk (beta) that they add to a diversified portfolio
  - ➔ Jensens’ $\alpha = Average\ Return - \beta_{value} \times MRP = 1.34\%$
  - More on this later.

Firm-Specific vs. Market Risk

A "General" Market Model

Market (Economy-Wide) and Firm-Specific Risk

- Denote:
  - $r_{i,t}$ = Rate of return on security i at time t.
  - $r_{m,t}$ = Market return at time t.
  - $\varepsilon_{i,t}$ = Specific return of firm i at time t.
- Divide the risk of an individual security into **economy-wide** risk and **firm-specific** risk. We do this formally in a regression framework as follows:

$$r_{i,t} = \alpha_i + \beta_i \times r_{m,t} + \varepsilon_{i,t}$$

- **Remarks:**
  - The larger is β_i, the more subject to market-risk this firm is.
  - The larger is $\varepsilon_{i,t}$, the more important firm-specific risk is.
  - This firm-specific risk is **independent** of market risk.

A “General” Market Model

- From statistics: if two variables are independent, then the variance of the sum equals the sum of the variances.
- The total variance (total risk) of i is:

$$\sigma_i^2 = \beta_i^2 \times \sigma_m^2 + \sigma_\varepsilon^2$$

- The **total risk** of a security is made up of the risk from market movements and the risk from firm-specific movements.
  - The first term is the variance of the market risk.
  - The second term is the variance of the idiosyncratic risk.

- **Bottom Line:** Two securities can have the **same** total risk, yet that risk can be made up of very **different** components.

A "General" Market Model

- Conceptually, the market return **r_m** is a measure of economy-wide returns, and it consists of all traded assets in the economy.
  - The market portfolio offers the **maximum** attainable diversification. It has no firm-specific risk.
- If an asset has **high** β, then it has large market risk.
  - The systematic variance (economy-wide risk) is $\beta_i^2 \sigma_m^2$.
- Instead, **any firm-specific risk can be diversified away**. Investors do not care about this type of risk!
  - The part that is not explained by the market return is σ_ε^2.
  - This part provides the unsystematic, firm-specific, or idiosyncratic component of the asset's return.
  - The more volatile ε is, the more important firm-specific risk is.

The CAPM is a "special case"

- In the CAPM, only the **systematic** component is priced!!!
  - Firm-specific risk can be eliminated by holding large portfolios.
  - Investors holding diversified portfolios are exposed only to systematic market-related risks.
  - Therefore, the relevant risk in the market's risk/expected return trade-off is market risk, **not** total risk.

- Investors are compensated **only** for holding assets with large amounts of market risk.
  - Holding that asset will add to the risk of their portfolios.

- **Bottom Line:** CAPM predicts that a security's return is related to the portion of the risk that cannot be eliminated by portfolio combination (ie: diversification).

The Security Market Line (SML)

- The CAPM can also be written as a **linear** relation between the β of a security and its expected rate of return.

$$E[r_i] - r_f = \beta_i \times [E(r_m) - r_f]$$

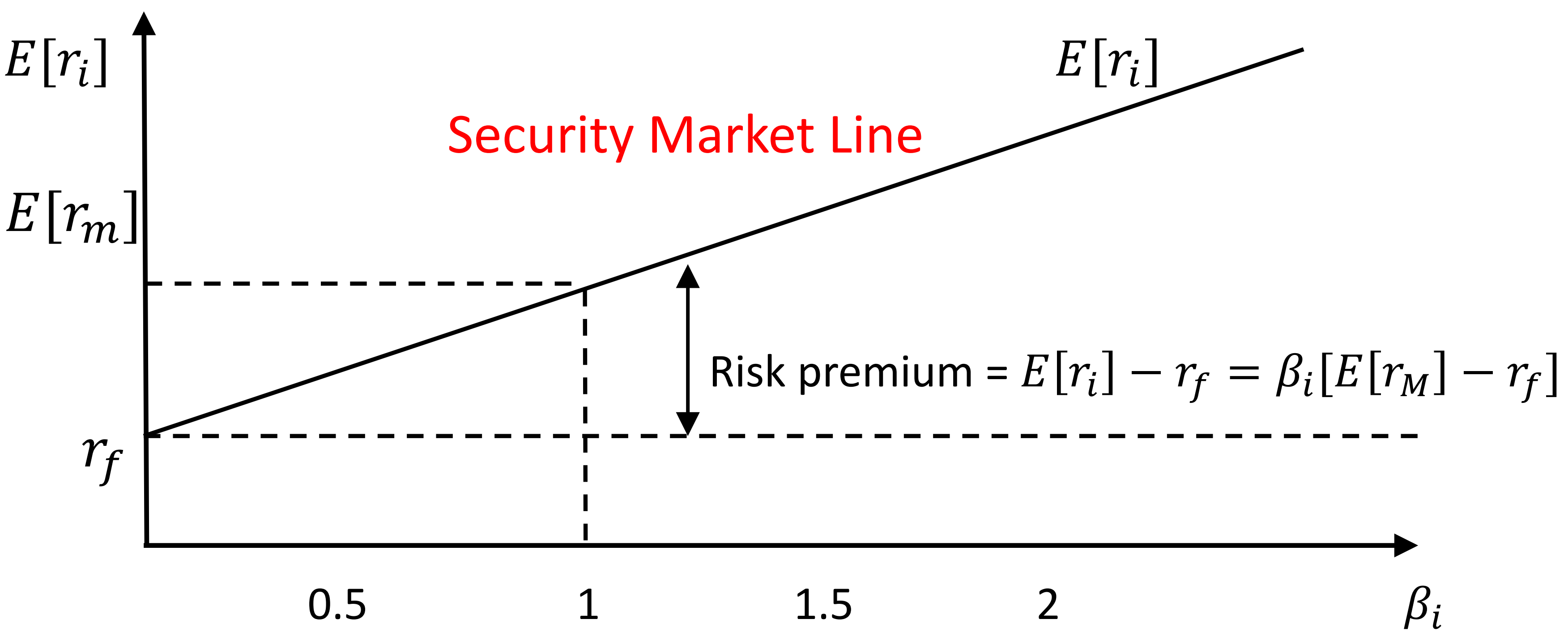
where

  - $E[r_i]$ = Expected rate of return on the security
  - $E[r_m]$ = Expected rate of return on the market portfolio
  - r_f = Risk-free rate

- **Bottom Line:** The relation is called the **security market line** (SML).
  - The expected risk premium varies in **direct** proportion to beta.
  - This means that all investments must plot along the sloping line, the security market line.

The Security Market Line (cont.)

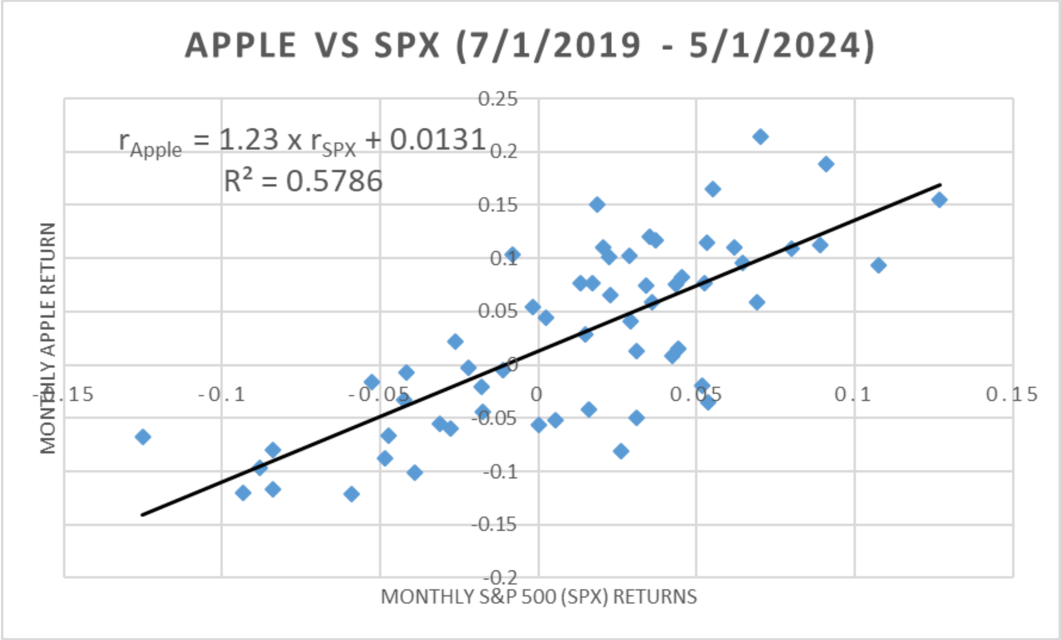
- The CAPM implies that **differences** in expected returns can be attributed entirely to differences in βs.
  - If you rank a group of assets according to their betas, their expected returns should fall on the SML.



Example I: Market Risk and R^2

- Suppose that in monthly data for Apple:
  - $\sigma^2_{Apple} = 0.00725$, $\sigma^2_m = 0.00276$, $\beta = 1.23$
- **Question:** What fraction of Apple's risk is explained by the market? What fraction of Apple's risk do you get paid to hold?
  - We are interested is in the **proportion** of market risk in total risk:

$$R^2 = \frac{\text{Systematic Risk}}{\text{Total Risk}} = \frac{\beta^2_{Apple} \times \sigma^2_M}{\sigma^2_{Apple}} = \frac{1.23^2 \times 0.00276}{0.00725} = 57.86\%$$



Example II: Decomposing Total Risk of a Stock

- Consider **two** stocks:

  A: An automobile firm: $\beta_A = 1.5$; $\sigma_{\varepsilon A}^2 = 0.10$

  B: An oil exploration firm: $\beta_B = 0.5$; $\sigma_{\varepsilon B}^2 = 0.18$

- The variance of the market-risk is measured by $\sigma_m^2 = 0.04$
- **Question:** What is the total risk of each stock? Which one will have a higher expected rate of return?

$$\sigma_A^2 = \beta_A^2 \sigma_m^2 + \sigma_{\varepsilon A}^2 = 1.5^2 \times 0.04 + 0.10 = 0.19$$

$$\sigma_B^2 = \beta_B^2 \sigma_m^2 + \sigma_{\varepsilon B}^2 = 0.5^2 \times 0.04 + 0.18 = 0.19$$

  - Therefore

$$R_A^2 = \frac{\beta_A^2 \times \sigma_m^2}{\sigma_A^2} = \frac{1.5^2 \times 0.04}{0.19} = 47\%$$

$$R_B^2 = \frac{\beta_B^2 \times \sigma_m^2}{\sigma_B^2} = \frac{0.5^2 \times 0.04}{0.19} = 5\%$$

Estimating Beta

Estimating Beta

- A standard way to estimate a beta is to use the **regression:**

$$E(r_i) - r_f = \alpha_i + \beta_i [E(r_m) - r_f] + \varepsilon_i$$

or

$$E(r_i) = \alpha_i + \beta_i r_m + \varepsilon_i$$

- We use historical monthly data to estimate beta.
  - Typically we use 5 years (60 months) of data.
    - Parameters tend to be stationary for up to 5 years.
  - Some institutions use more than 5 years.
  - Some institutions use more frequent data (weekly, daily)

Estimating Beta: Merck

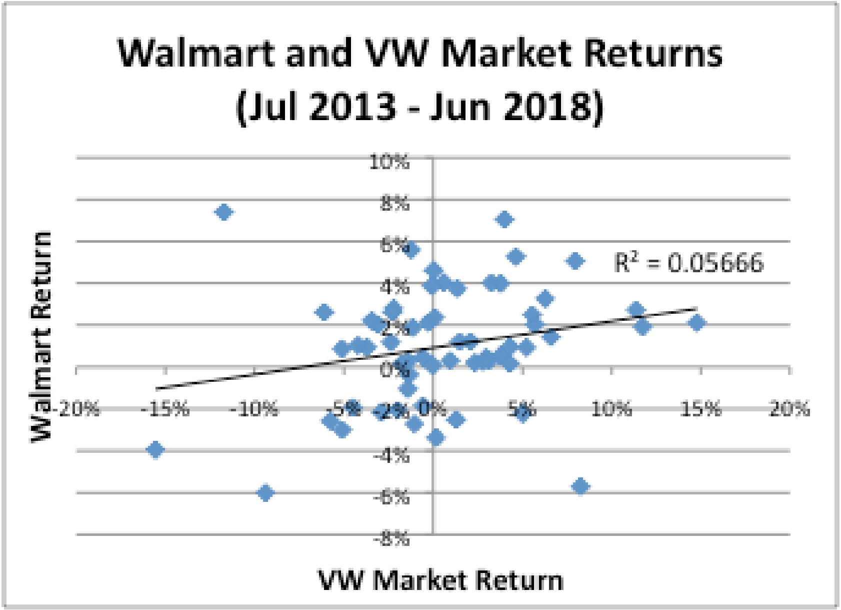
- Here is an example of such a regression for Merck common stock using 5 years of monthly data from 2019 to 2024

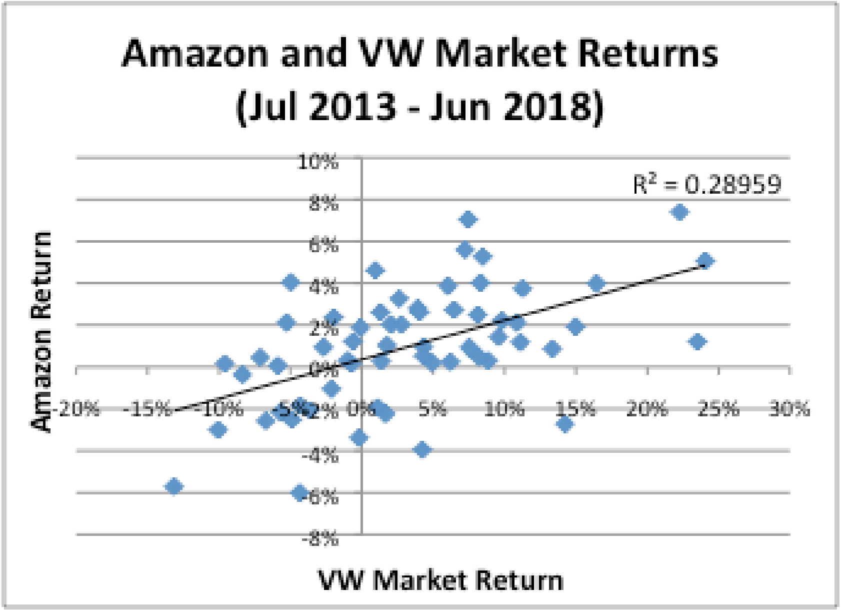
Regression Analysis: CAPM				
	Excess Returns		**Simple Returns**	
	alpha	**beta**	**alpha**	**beta**
Estimates	0.0071	0.4140	0.0081	0.4158
Standard Errors	0.0076	0.1441	0.0077	0.1441
R squared	0.1265		0.1122	

- **Question:** What is the expected return for Merck equity holders?
  - For a market risk-premium of $E(r_m) - r_f = 7\%$ and a $r_f = 5.47\%$, using CAPM we obtain:

$$E[r_i] = r_f + \beta_i \, E(r_m) - r_f] = 5.47\% + 0.4140 \times 7\% = \mathbf{8.37\%}$$

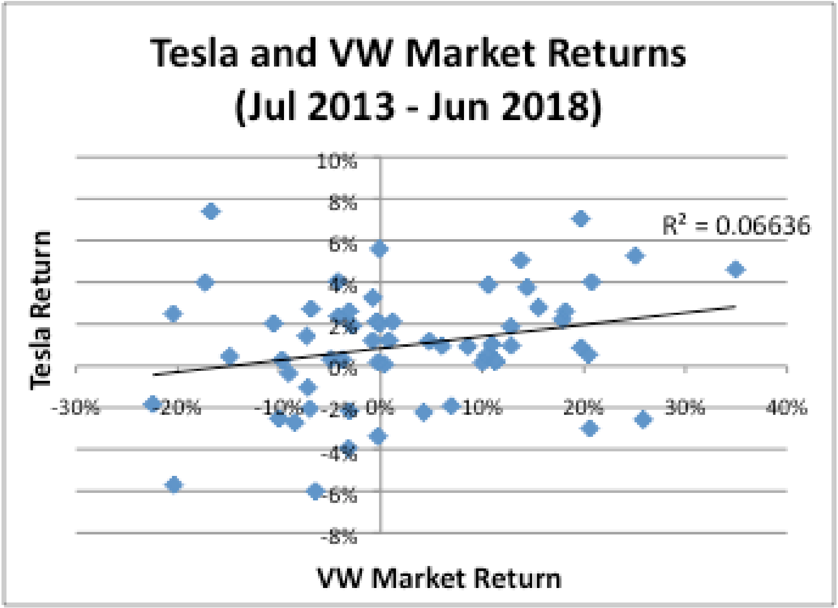
Walmart, Amazon, and Tesla





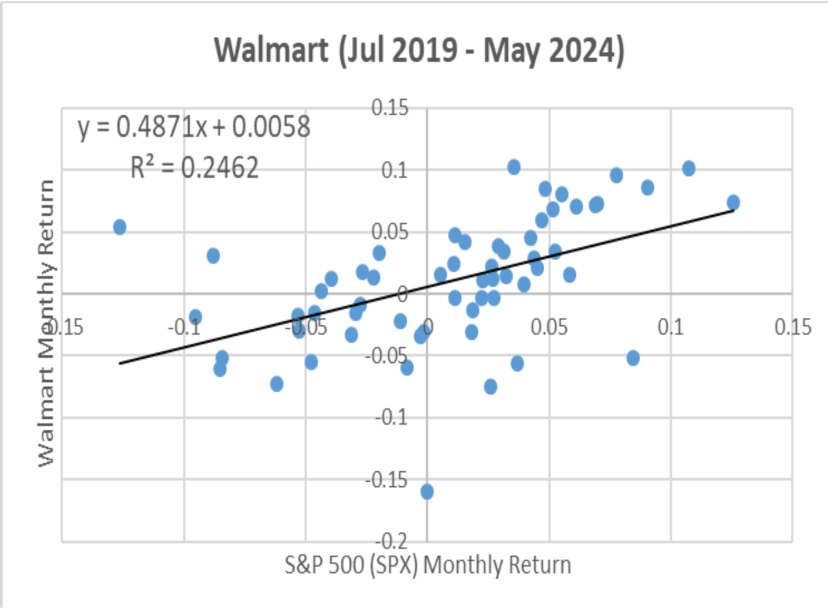
$R^2 = 0.06$; $\beta = 0.44$

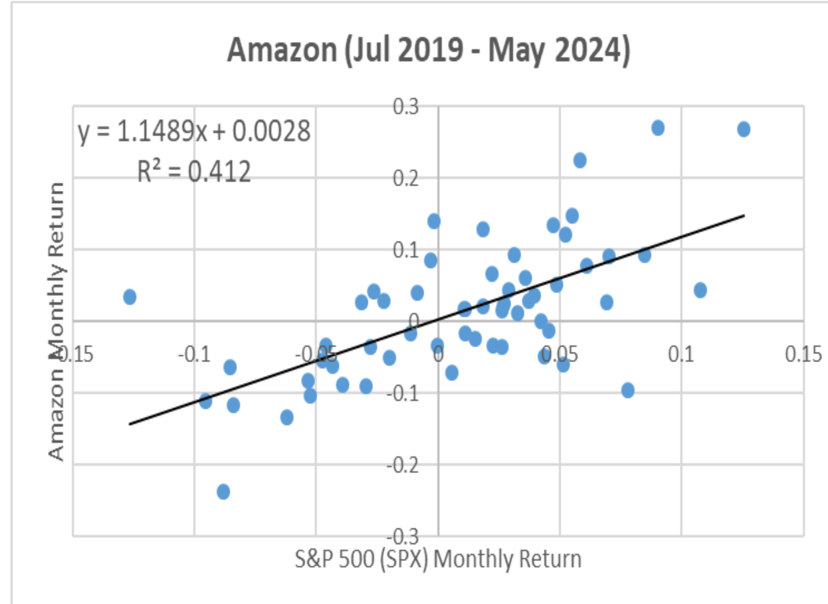
$R^2 = 0.29$; $\beta = 1.53$



$R^2 = 0.07$; $\beta = 1.15$

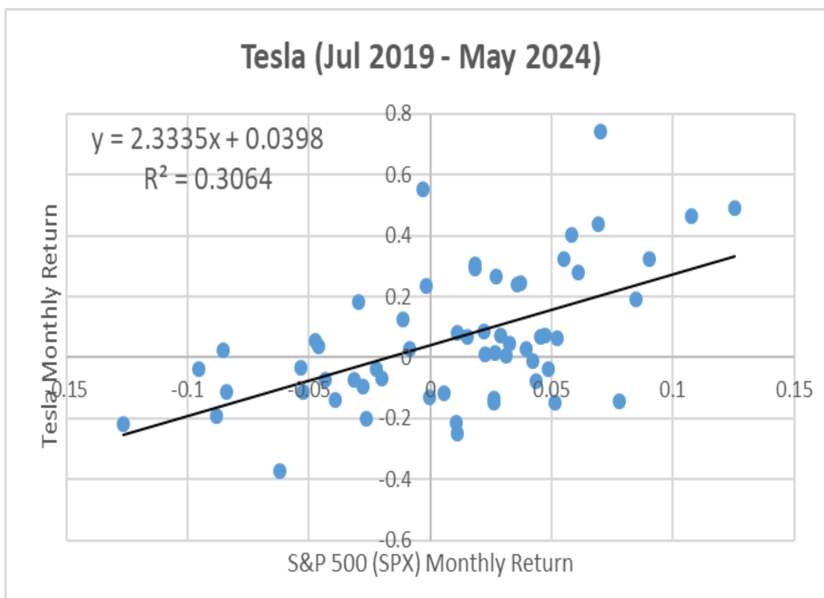
Walmart, Amazon, and Tesla (Update)





$R^2 = 0.25$; $\beta = 0.48$

$R^2 = 0.41$; $\beta = 1.15$



$R^2 = 0.31$; $\beta = 2.33$

Empirical Evidence

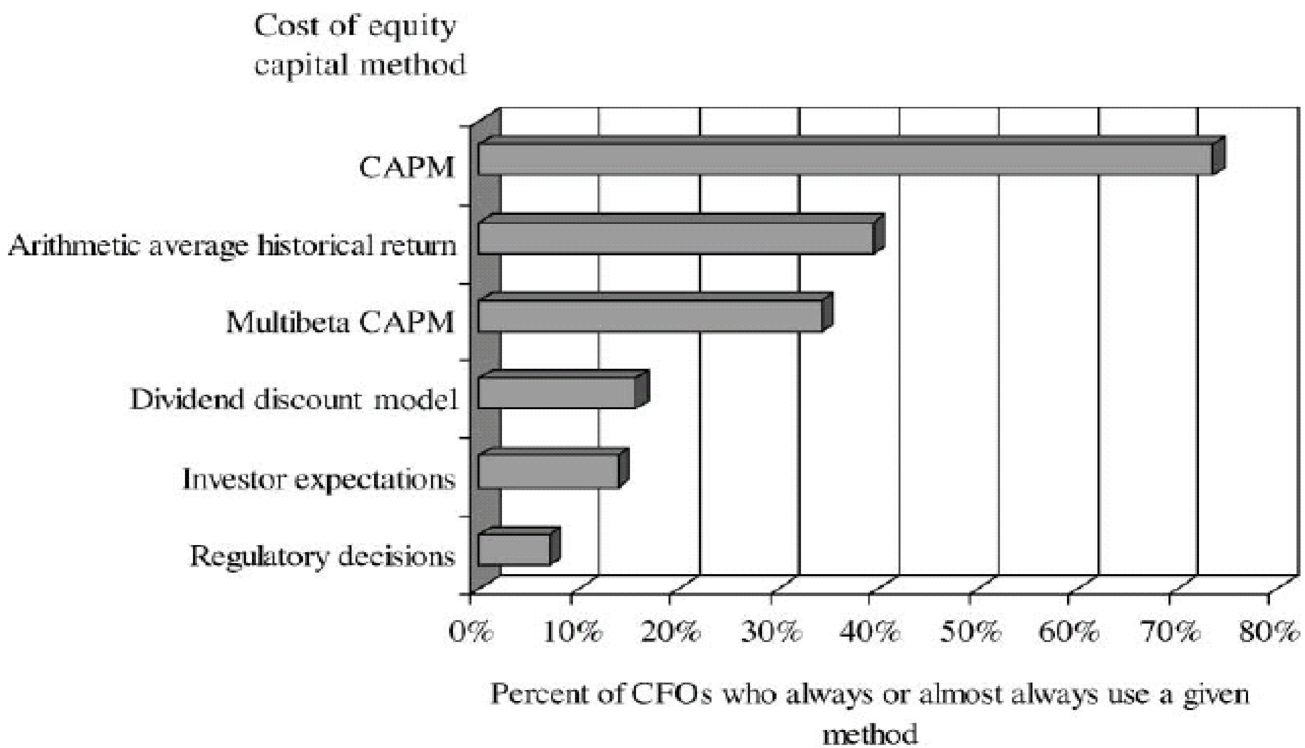
Tests of the CAPM: Criticisms

- **Criticism I:** Get different results in different time periods.
  - Difficult to explain the reasons
  - The "noise" makes it impossible to judge whether the model holds better in one period than in another.

- **Criticism II:** Returns have been related to **other** measures.
  - Some types of assets do better than their betas would suggest:
    - Firms with low betas do better than predicted by CAPM.
    - Small firms had higher returns than predicted by CAPM.
    - High book/market stocks had better returns than low book/market stocks with similar betas.

- **Criticism III:** The market portfolio should contain **all** risky investments, including stocks, bonds, commodities, real estate.
  - Most market indexes contain only a sample of common stocks.

Should We Abandon The CAPM Then? No!



Graham, J. and Campbell Harvey, *Journal of Financial Economics*, 2001

- Can use other models, but beta from the CAPM is still the most commonly used and has a measure of risk that can be traced back to some foundations
  - Many of the alternative factors that explain returns may be a result of data mining.

Takeaways

- Total risk is defined as variability in return.
  - The investor can reduce risk by holding a diversified portfolio.

- The total risk of a security can be divided into unsystematic and systematic risk:
  - Risk that can be eliminated through **diversification** is called **idiosyncratic risk**. It is associated with the events unique to the firm and independent of other firms.
  - The risk remaining in a diversified portfolio is called systematic risk.

- If CAPM correctly describes market behavior, investors hold diversified portfolios to minimize risk.
  - Since investors hold diversified portfolios with the CAPM, they are only compensated for systematic risk.

Takeaways (cont.)

- The risk of a well-diversified portfolio depends only upon the economy-wide risk of the securities included in the portfolio. This is measured by the average **β**.

- The risk/expected return trade-off with the CAPM is called the Security Market Line (SML).
  - Securities are prices such that:

$$E(r_i) = r_f + \beta_i[E(r_m) - r_f] \rightarrow E(r_i) - r_f = \beta_i [E(r_m) - r_f]$$

Appendix: More on CAPM

CAPM: Main Assumptions

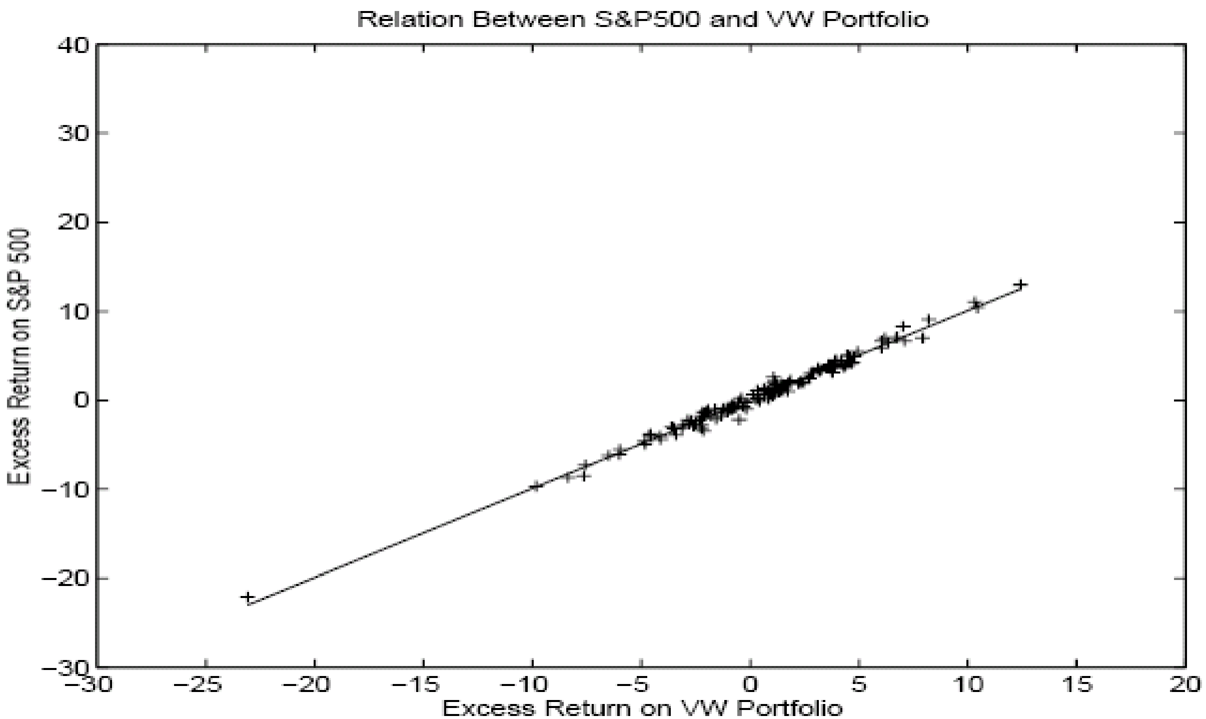
- We assume that investment in T-Bills is risk-free.
- Investors can borrow money at the same rate of interest at which they can lend.
  - In reality, borrowing rates are generally higher than lending rates.
- Investors are mean-variance investors: they choose portfolios based on expected returns and standard deviations.
- All investors have the same information: everyone faces the same MVF (Minimum Variance Frontier).
  - Investors are content to invest their money in a limited number of benchmark portfolios: the risk-free portfolio and the market portfolio.

CAPM: Technical Details

- Investors like high expected returns and low standard deviation.
  - Common stock portfolios that offer the highest E[r] for a given σ are known as efficient portfolios.
- The investor can lend or borrow at the risk-free interest rate.
- The composition of this best efficient portfolio depends on the investor's assessments of E[r], σ, and ρ.
  - If there is no superior information, each investor should hold the same portfolio as everybody else.
  - In other words, everyone should hold the market portfolio.
- A stock's sensitivity to changes in the value of the market portfolio is known as **beta**.
  - Beta measures the marginal contribution of a stock to the risk of the market portfolio.

More on S&P 500 returns…

- To get the S&P 500 return each day, S&P multiplies the weights for each of the 500 stocks by the stock returns.
  - The weights change each day as relative stock prices rise and fall.
- The S&P 500 index return is a **good proxy** for the overall US equity market return since it is very highly correlated with the rest of the stock market.



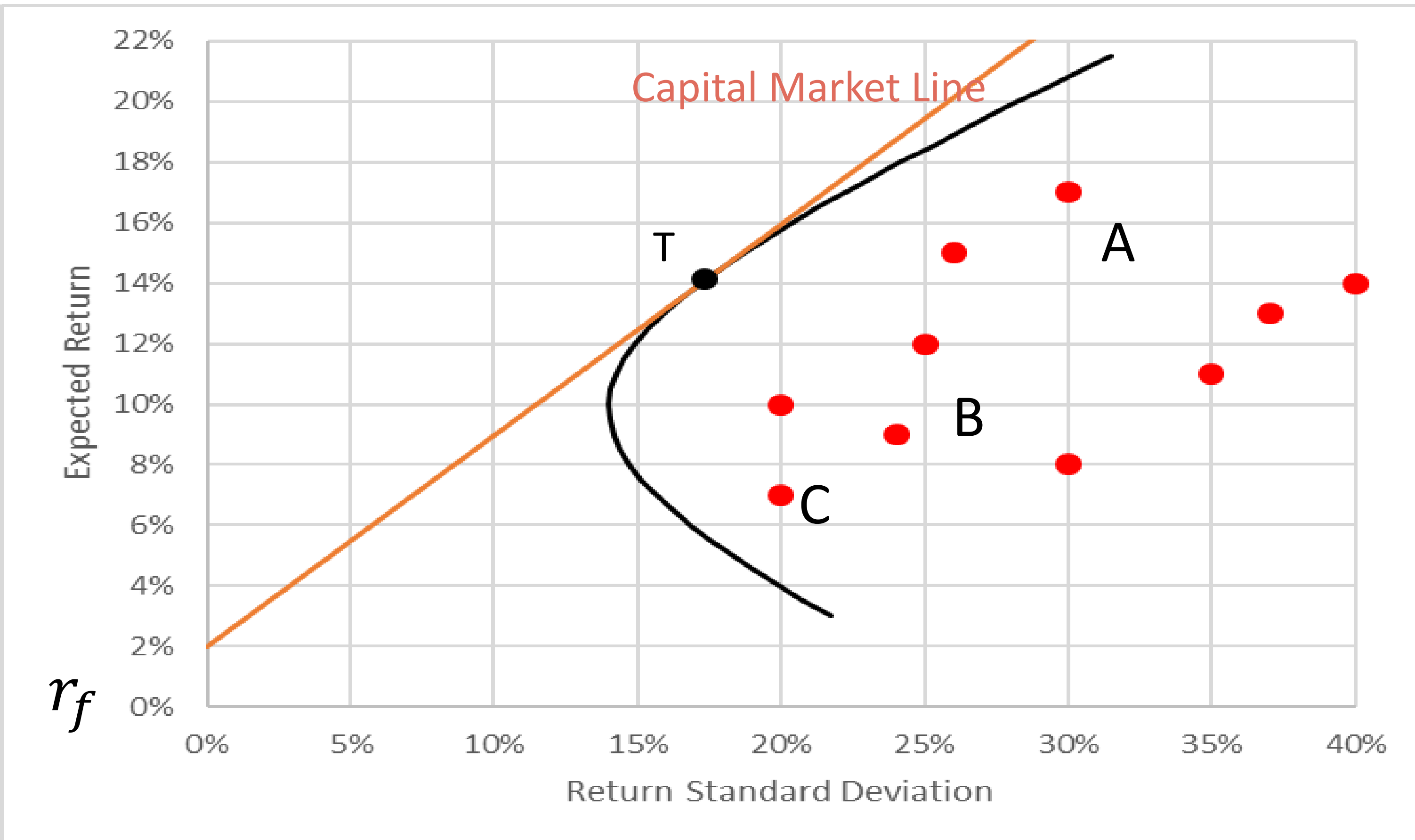
Corr(S&P500,VW) = 0.99

$\beta_{S\&P\ 500,\ VW} = 1$

Appendix: Derivation of CAPM

Back to Tangency Portfolio....

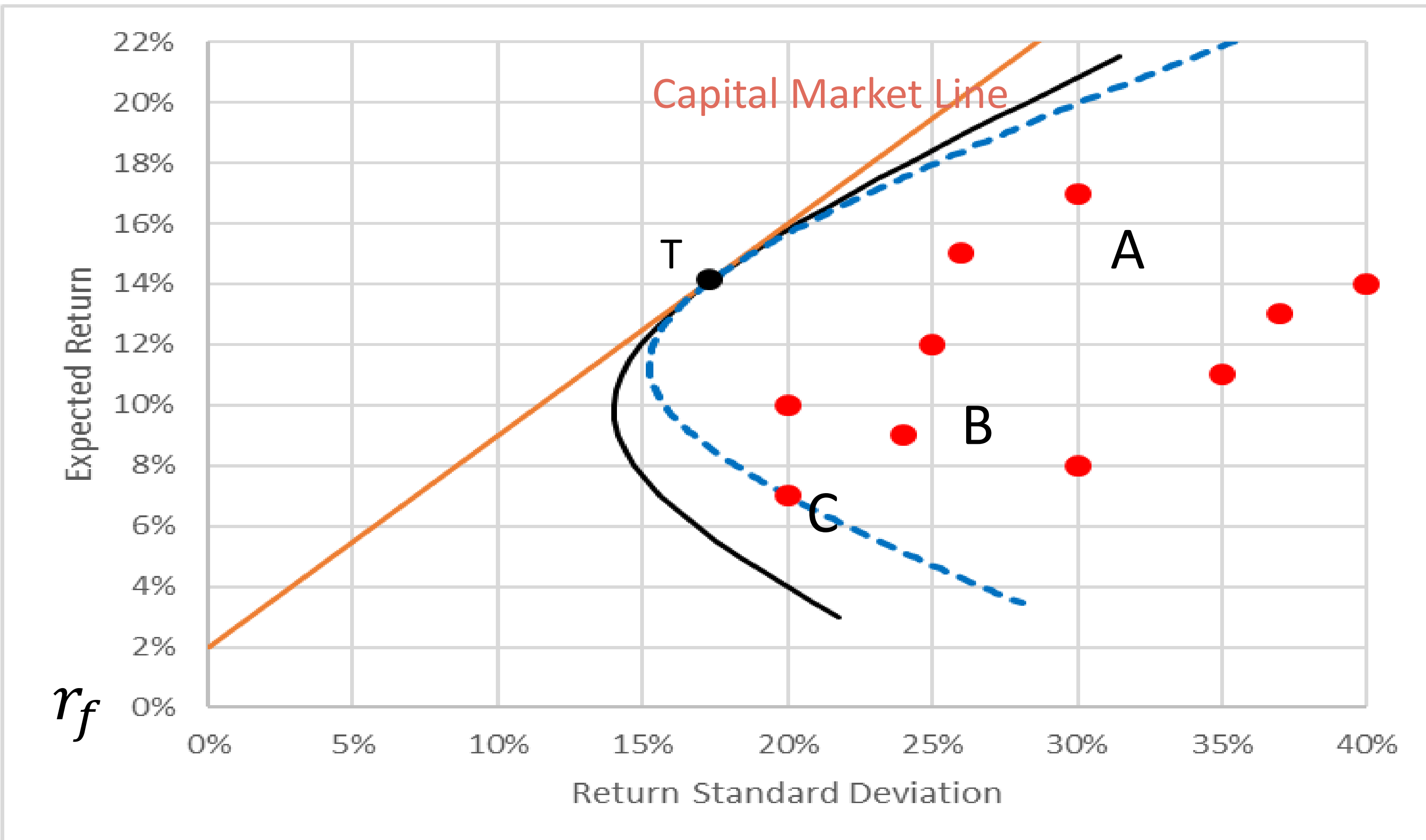
- Recall the **minimum-variance frontier (MVF)** and the Tangency portfolio (T)
  - Capital Market Line = Straight line from r_f tangent to T



T is a Super Tangency Portfolio

- The Tangency portfolio T is in fact a "**Super"-Tangency portfolio**.
  - Mixing T with any asset (A, B, C,….)➔ the arc is also tangent to MVF.

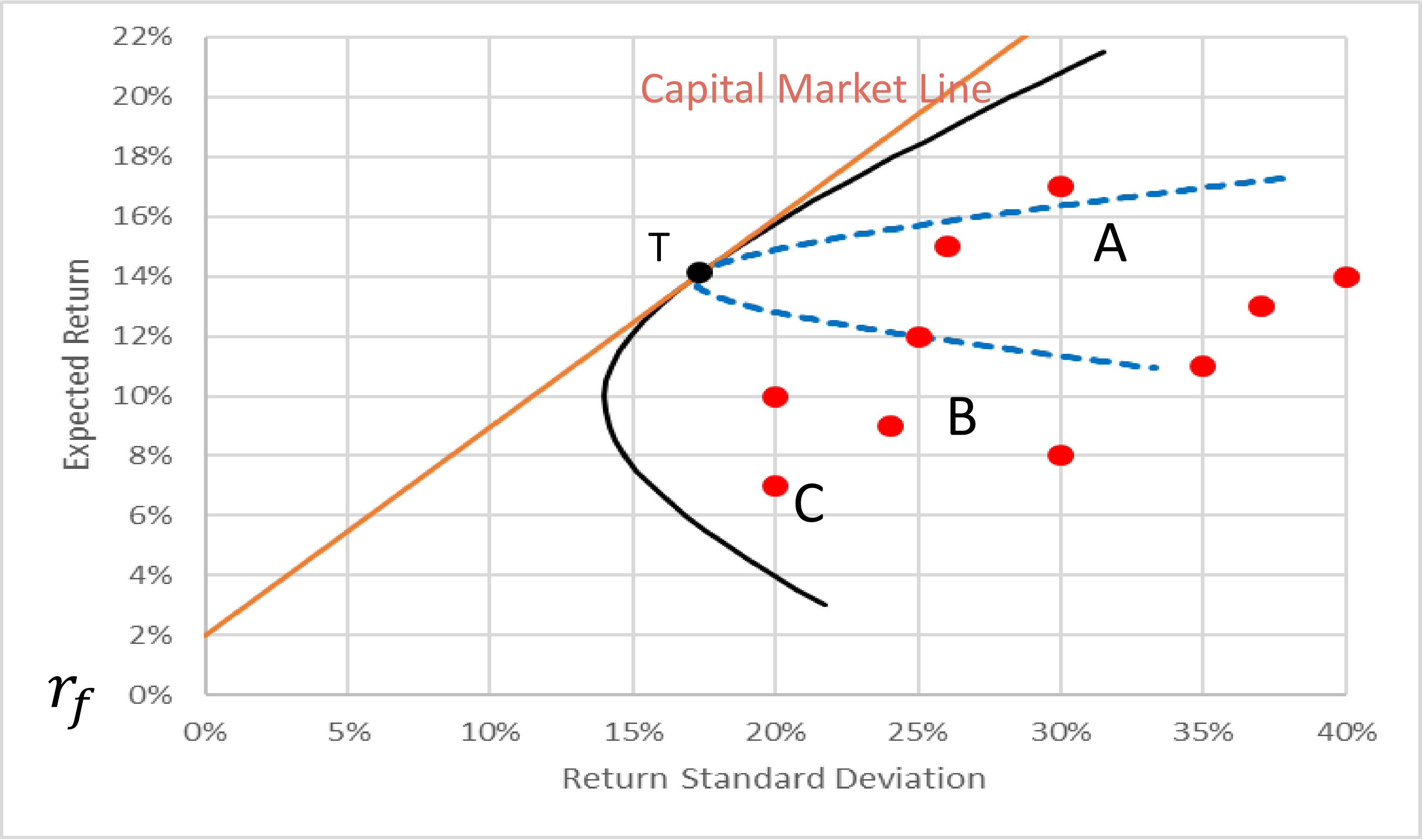
Tangency of **T - C** with MVF and CML



T is a Super Tangency Portfolio

- The Tangency portfolio T is a **Super-Tangency portfolio**.
  - Mixing T with any asset (A, B, C,....)➔ the arc is also tangent to MVF.

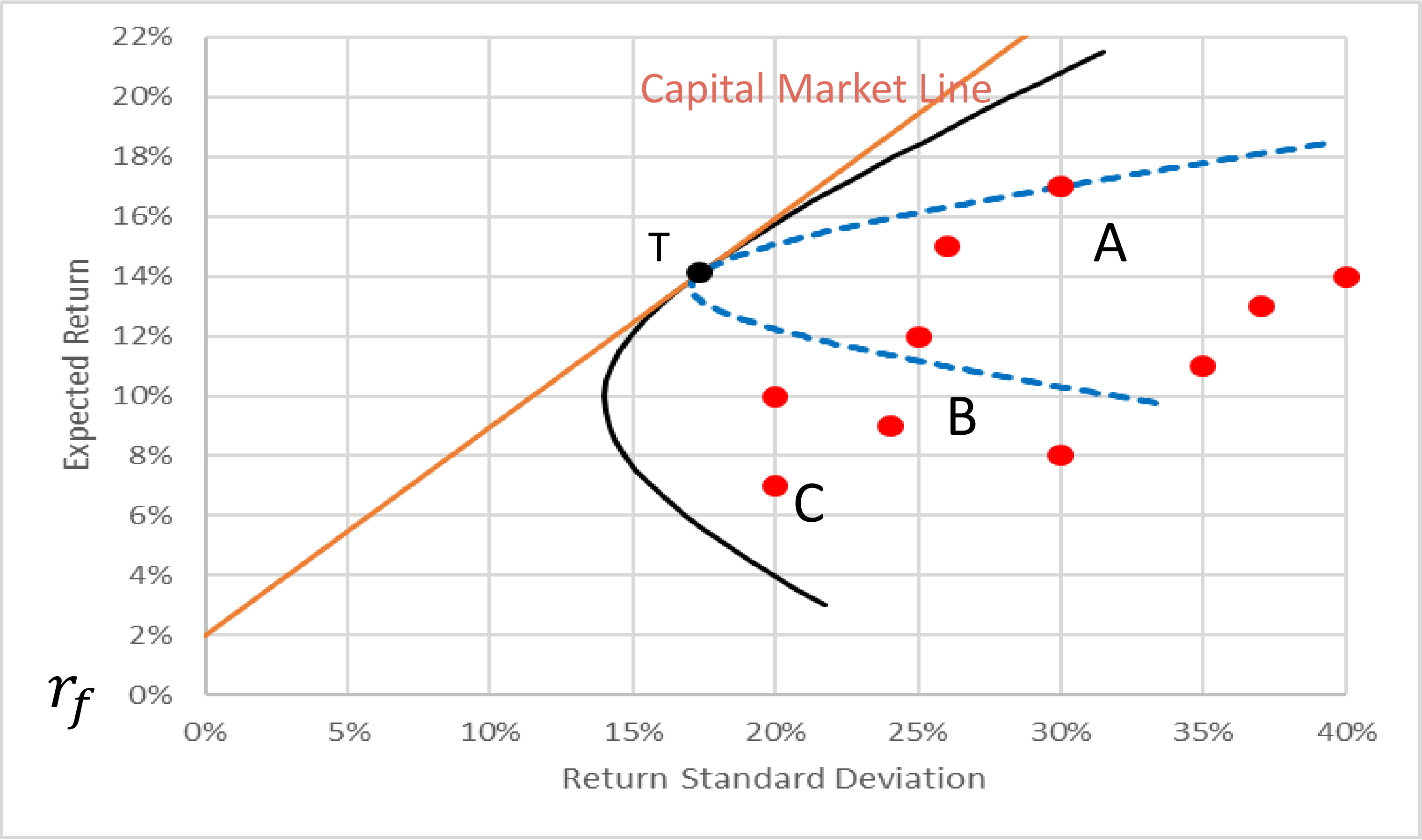
Tangency of **T - B** with MVF and CML



T is a Super Tangency Portfolio

- The Tangency portfolio T is a **Super-Tangency portfolio**.
  - Mixing T with any asset (A, B, C,….)➔ the arc is also tangent to MVF.

Tangency of **T - A** with MVF and CML



Bottom Line

1. In equilibrium, **Tangency portfolio (T) = Market portfolio (M)**
2. From the Market M (=T), I can shift a small amount x either:
   a) into risk-free rate ➔Travel along Capital Market Line; or
   b) into any risky asset (say A) ➔ Travel along the arc T - A
3. For small x➔ impact on reward/risk ratio for (a) and (b) is the same due to super-tangency (see figure).
   a) Slope of Capital Market Line = Sharpe Ratio = $\frac{E(r_m)-r_f}{\sigma_m}$
   b) Slope on arc from T to A is = $\frac{E(r_p)-E(r_m)}{\sigma_p-\sigma_m}$
4. It follows:

$$\frac{E(r_p)-E(r_m)}{\sigma_p-\sigma_m} = \frac{E(r_m)-r_f}{\sigma_m} \qquad (1)$$

5. Algebra shows Eq. (1) equals $E(r_A) = r_f + \beta_A\,[E(r_m) - r_f]$

Quick Derivation of CAPM (for the curious)

- Here is a quick derivation of CAPM:
  - Shift x from M to A ➔ $r_p = x\, r_A + (1-x) r_m$
  - Change in Expected Return: $E(r_p) - E(r_m) = x[E(r_A) - E(r_m)]$
  - Change in Risk (turns out): $\sigma_p - \sigma_m \approx x\, \sigma_m\, [\beta_A - 1]$
    - As we saw, if $\beta_A > 1$ ➔ $\sigma_p > \sigma_m$
    - if $\beta_A < 1$ ➔ $\sigma_p < \sigma_m$
  - Therefore (see figure):

$$Slope\ of\ Arc = \frac{E(r_p) - E(r_m)}{\sigma_p - \sigma_m} = \frac{E(r_m) - r_f}{\sigma_m} = Slope\ of\ CML$$

$$\Rightarrow \frac{x[E(r_A) - E(r_m)]}{x\, \sigma_m\, [\beta_A - 1]} = \frac{E(r_m) - r_f}{\sigma_m}$$

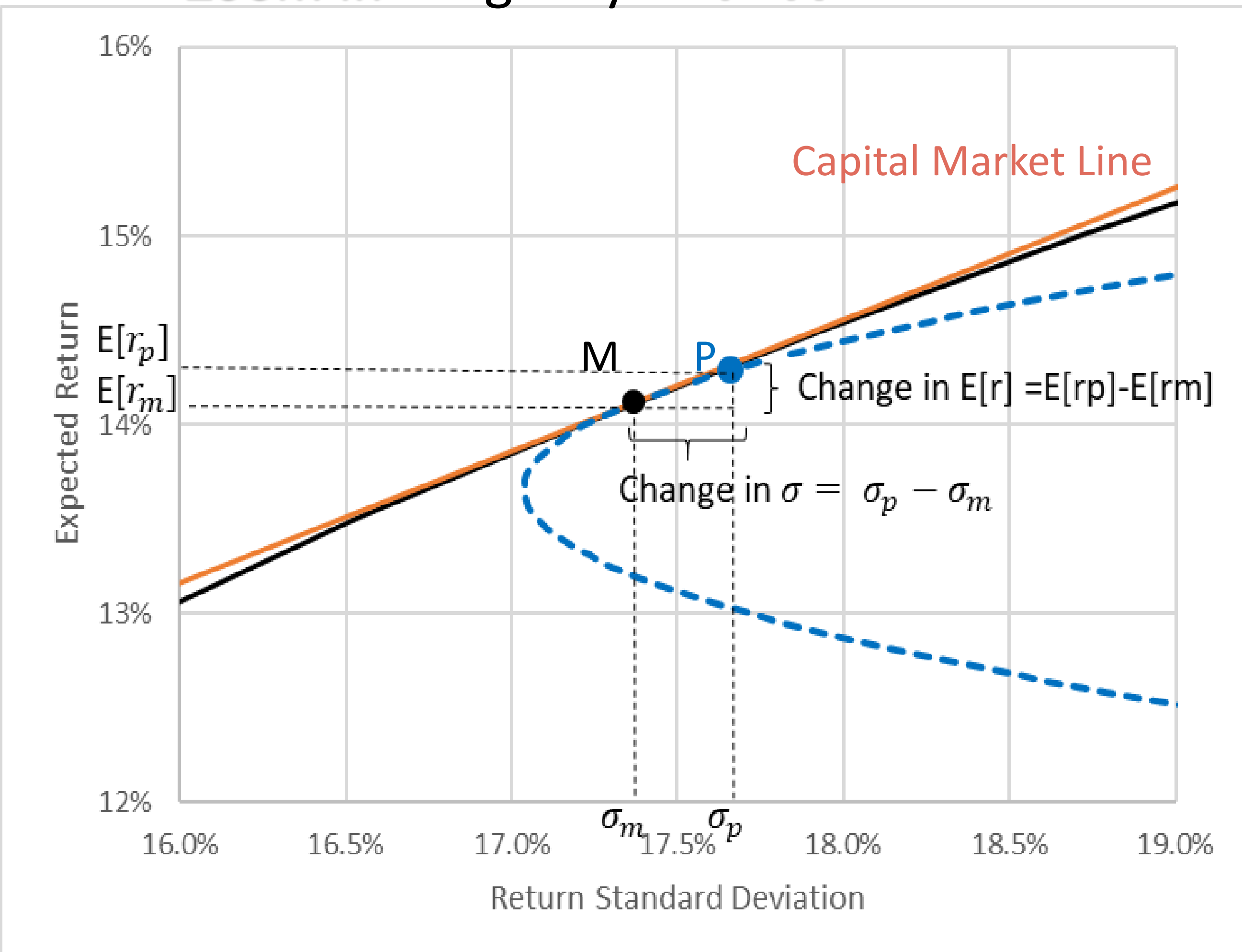
$$\Rightarrow E(r_A) - E(r_m) = [\beta_A - 1][E(r_m) - r_f]$$

$$\Rightarrow E(r_A) - r_f = \beta_A [E(r_m) - r_f]$$

T is a Super Tangency Portfolio

- The Tangency portfolio T is a **Super-Tangency portfolio**.
  - Mixing T with any asset (A, B, C,….)➔ the arc is also tangent to MVF.

Zoom in Tangency of **T - A** with MVF and CML



Slope of CML = $\frac{E(r_m)-r_f}{\sigma_m}$

Slope of Arc = $\frac{E(r_p)-E(r_m)}{\sigma_p-\sigma_m}$

One more Step in the Derivation of CAPM

- Here is a quick derivation of CAPM:
  - Shift x from M to A ➔ $r_p = x\, r_A + (1-x) r_m$
  - Change in Expected Return: $E(r_p) - E(r_m) = x[E(r_A) - E(r_m)]$
  - Change in Risk (all worked out)

$$\sigma_p^2 = x^2\sigma_A^2 + (1-x)^2\sigma_m^2 + 2x\,(1-x)\mathrm{cov}(r_A, r_m)$$
$$= x^2\sigma_A^2 + (1-2x+x^2)\sigma_m^2 + (2x-2x^2)\mathrm{cov}(r_A, r_m)$$

- For x small, $x^2 \approx 0$ ➔ $\sigma_p^2 \approx \sigma_m^2 - 2x\,\sigma_m^2 + 2x\ \mathrm{cov}(r_A, r_m)$
- Considering σ_p^2 as a function of x, $\sigma_p^2(x)$, with $\sigma_p^2(0) = \sigma_m^2$, the first-order Taylor expansion (for x small) is:

$$\sigma_p(x) = (\sigma_p^2(x))^{\frac{1}{2}} \approx \left(\sigma_p^2(0)\right)^{\frac{1}{2}} + \frac{1}{2}\left(\sigma_p^2(0)\right)^{-\frac{1}{2}} \frac{d\sigma_p^2(x)}{dx}\ x$$
$$\approx \sigma_m + \frac{1}{2\sigma_m}\left(-2\,\sigma_m^2 + 2\ \mathrm{cov}(r_A, r_m)\right)x$$

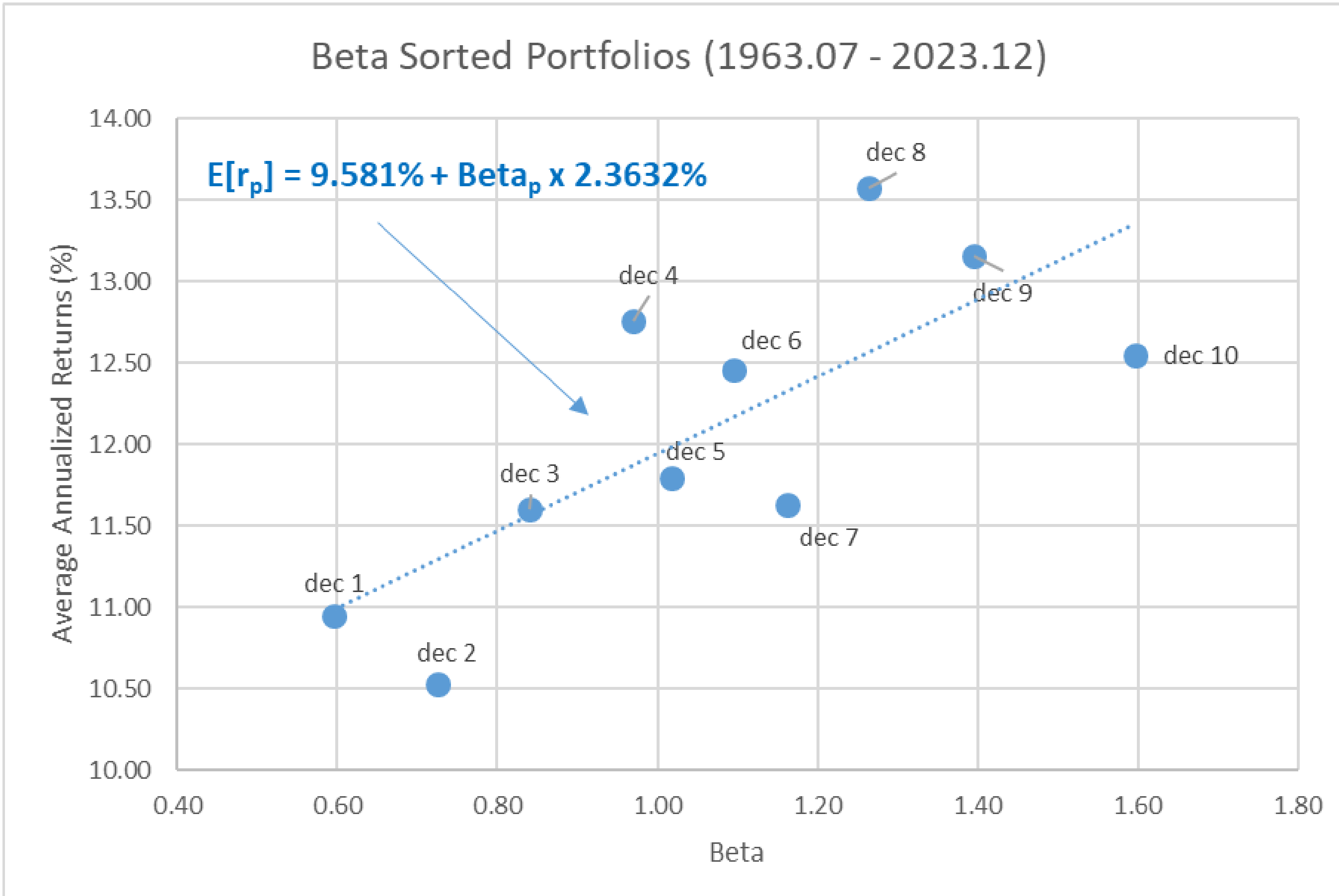
➔ $\boldsymbol{\sigma_p(x) - \sigma_m \approx \sigma_m\left(-1 + \frac{\mathrm{cov}(r_A, r_m)}{\sigma_m^2}\right)x}$

- As we saw, if $\beta_A > 1$ ➔ $\sigma_p > \sigma_m$ and if $\beta_A < 1$ ➔ $\sigma_p < \sigma_m$

Test of the CAPM: Do Higher Beta Stocks Fetch Higher Average Returns?

- Every year (1963.07 – 2023.12):
  - Rank stocks on their past beta, put them in decile portfolios
  - Compute average return of portfolio over the next year

YES! Positive relation between betas and average *future* returns

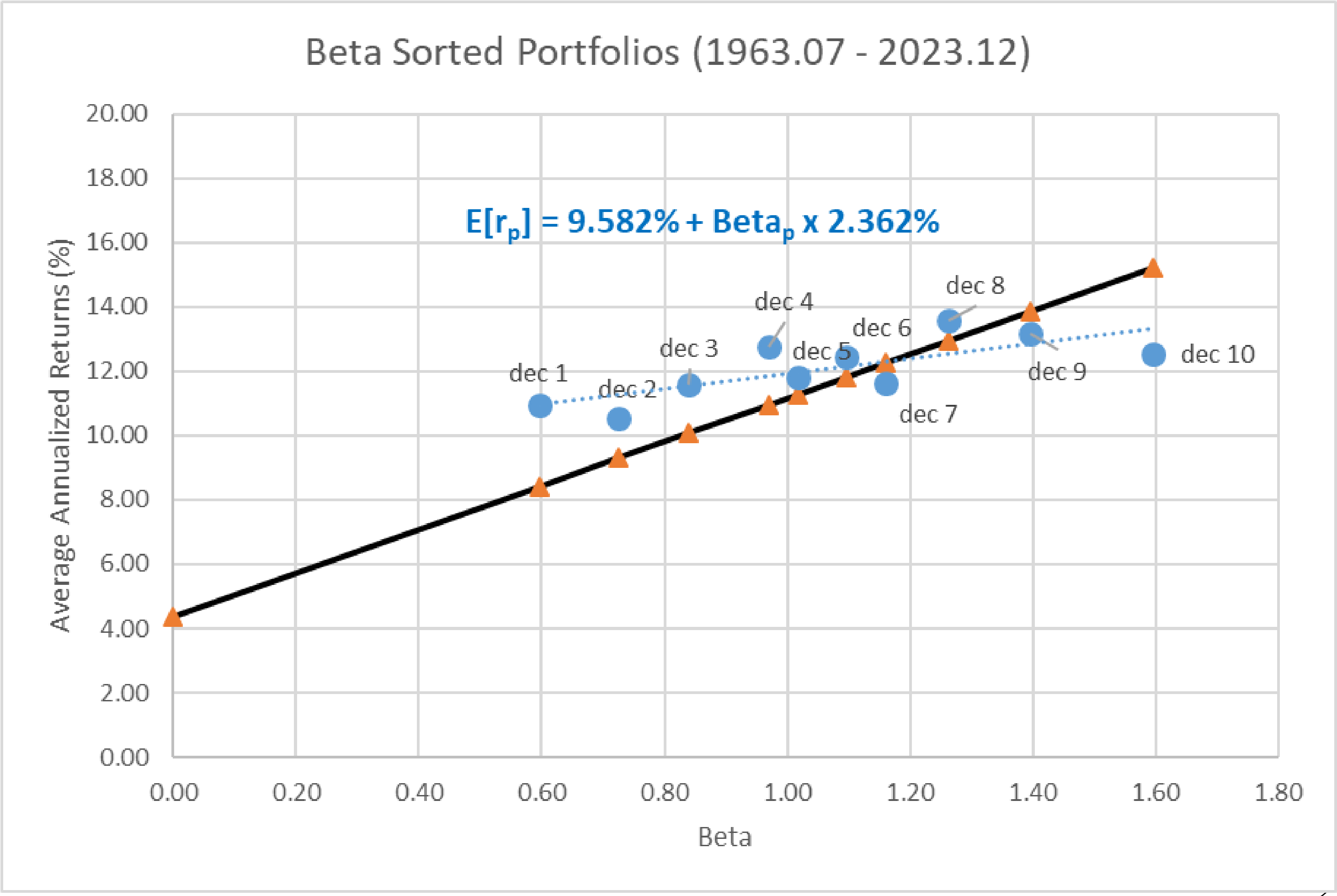


Test of the CAPM: Do Higher Beta Stocks Fetch Higher Average Returns?

- **Do they lie on the Security Market Line?**

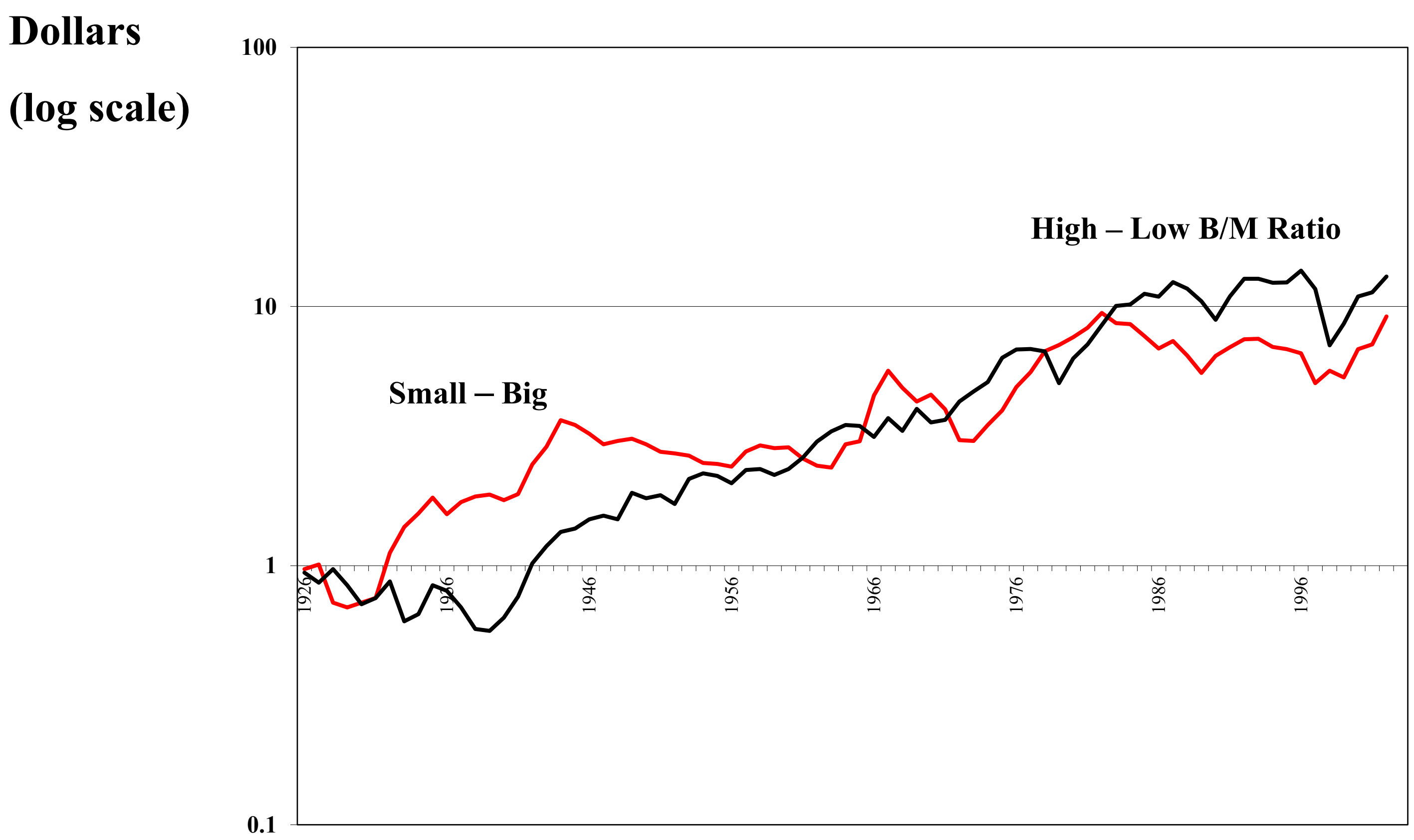
$$E[r_p] = r_f + \beta_p \times MRP \rightarrow E[r_p] = 4.35\% + \beta_p \times 6.81\%$$

NO! The slope should be the MRP (6.81%), but it is flatter (2.36%)



Tests of the CAPM (cont.)

Firm Size and Book-to-Market predict returns not captured by Beta



http://mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html

Some Alternative Theories

- The CAPM has come under attack for its inability to sufficiently explain the empirically observed differences in returns across different securities.
  - The CAPM is sometimes called the market model or **single factor model**.
- Researchers have found that factors other than the return on the market help predict returns.
  - This has led to the development of a multi-factor linear model called the **Arbitrage Pricing Theory (APT).**

Some Alternative Theories: APT

- The Arbitrage Pricing Theory (APT) attempts to account for other risk factors:
  - It does not ask which portfolios are efficient. It starts by assuming that each stock's return depends partly on pervasive macroeconomic influences or "factors".
  - A standard empirical implementation includes industrial activity, inflation, and interest rate factors (among others).

$$r_i = \alpha_i + \beta_{i1}Z_1 + \beta_{i2}Z_2 + \beta_{i3}Z_3 + \ldots + \varepsilon_i$$

  - In the APT, each stock has a different β on each factor.
  - Estimates of expected returns based on the APT are also available commercially.

Some Alternative Theories:
Fama-French Three-Factor Model

- Fama and French (1992) find that, as an empirical matter, expected returns can be explained by a **three-factor model**, where the factors are related to:
  - Market: $(r_{S\&P,t} - r_{f,t})$
  - Size: $(r_{Small,t} - r_{Big,t})$
  - Book-to-Market Ratio: $(r_{High,t} - r_{Low,t})$

	Fama-French Three Factor Model				
	Factor Sensitivities				**CAPM**
	β_{market}	β_{size}	$\beta_{book\text{-}to\text{-}market}$	Expected Risk Premium*	Expected Risk Premium
Aircraft	1.15	0.51	0.00	7.54%	6.43%
Banks	1.13	0.13	0.35	8.08%	5.55%
Chemicals	1.13	-0.03	0.17	6.58%	5.57%
Computers	0.90	0.17	-0.49	2.49%	5.29%
Construction	1.21	0.21	-0.09	6.42%	6.52%
Food	0.88	-0.07	-0.03	4.09%	4.44%
Petroleum & Gas	0.96	-0.35	0.21	4.93%	4.32%
Pharmaceuticals	0.84	-0.25	-0.63	0.09%	4.71%
Tobacco	0.86	-0.04	0.24	5.56%	4.08%
Utilities	0.79	-0.20	0.38	5.41%	3.39%

*The expected risk premium equals the factor sensitivities multiplied by the factor risk premiums, that is, $(\beta_{market} * 5.2) + (\beta_{size} * 3.2) + (\beta_{book\text{-}to\text{-}market} * 5.4)$.